

KuantorFlow — Weekly Development Report

Period: 14–17 July 2026 · Repositories: kuantorflow, ai_agent, kuantorflow_automation

Executive Summary

This week was the **Settings week**: the long-planned per-user settings platform went from an open ticket to a fully shipped feature chain — storage layer, Settings popup, applied behaviour, and real alternative providers — plus a long-standing duplicate-cards bug fixed at the root. **10 pull requests merged** across two repositories, and the regression suite grew from 42 to **72 passing tests** (after first being repaired from a broken state).

What a user sees today that they didn't on Monday: a **Settings** item in the header opens a popup where they can turn on *Add cards automatically*, choose **Google or Bing** as the translator, and **Oxford Learner's or Merriam-Webster** as the explanatory dictionary — each signed-in Google user with their own persistent config file, anonymous visitors sharing a default one. And looking up the same word twice no longer piles up duplicate cards: the app says "Already in DB" instead.

Repository	Role this week	Merged PRs
kuantorflow	Settings platform, providers, duplicate fix, SM-2 write-up	5
kuantorflow_automation	Suite repair, 30 new tests, gate helper, dedup script, deck tooling	5
ai_agent	No changes this week	0
Total		10

Per the report request, commit authorship was checked across all repositories for the period: every commit was authored by Kuantor (with Claude as co-author); no GitHub Copilot commits were found in this time range.

Completed Work by Theme

1. The settings storage layer (kuantorflow #97 → issue #86)

The foundation everything else stands on: per-identity JSON config files in `settings/` — `config-default.json` shared by anonymous visitors, `config-<username>.json` per Google-authorised user (email prefix, sanitised with the same rule as the per-user log directories).

- **Robust by design**: reads never raise (missing/corrupt files fall back to defaults), values are validated both ways (unknown keys dropped, bad values reset), writes are atomic (temp file + `os.replace`), and crafted emails can't escape the settings directory.

- **Self-provisioning** (#105): the first read for an identity now writes its config file with the defaults — nothing to provision on deploy. Corrupt files are never overwritten, so a hand-edited file stays inspectable.
- Adding a future setting = one entry in `settings_store.DEFAULTS`; existing files pick up the new default on their next read, no migration.

2. The Settings popup and applied behaviour (kquantorflow #105 → issues #13, #20)

A new **Settings** item in the header (between Home and About) opens a popup dialog, pre-filled server-side from the identity's config and saved through a validating POST `/settings` endpoint:

- **Add cards automatically** (#13, off by default) — when on, *Look up & save* writes parsed cards straight to the database with the green confirmation banner; when off, the review-before-save popup opens as before. (.mht uploads always go through review — editing the parsed lines is the point of that popup.)
- **Translation (ENG → UKR/RUS)** (#20) — Google Translate or Bing Translator.
- **Explanatory dictionary (ENG → ENG)** (#20) — Oxford Learner's Dictionaries (default) or Merriam-Webster.
- The lookup panel title follows the chosen translator, and the popup states whose settings are being edited (the signed-in email, or "all anonymous visitors").

3. Real providers behind the radio buttons (kquantorflow #106 → issue #21)

Lookups dispatch through `parsers.lookup_word(word, topic, translator, explanatory_dictionary)` to one fetcher per provider — all real implementations, verified live:

- **Bing Translator** — the week's most interesting engineering detour. `bing.com`'s own web endpoints turned out to **reject non-browser TLS clients**: replaying a real browser session's cookies, token, and exact request body from Python still returned 401, while the identical request from inside the browser returned 200. The solution reaches the same Microsoft Translator engine the way the Edge browser's built-in translate does — a short-lived anonymous JWT from `edge.microsoft.com`, then the official dictionary/lookup API — returning byte-identical translations to the `bing.com` page, from plain requests.
- **Oxford Learner's Dictionaries** — one entry page covers one part of speech (`run_1` verb, `run_2` noun), so sibling entries linked from the page's *Other results* box are fetched too (capped at 3 pages, lookalikes like `run-up_1` ignored).
- **Merriam-Webster** — a single page carries every part-of-speech entry.
- **Graceful degradation throughout**: a translator that fails or returns nothing falls back to Google; a dictionary that fails falls back to Reverso; definition failures never break a lookup.

4. Duplicate cards fixed at the root (kquantorflow #107 → issue #101)

Looking up a word twice used to add it to the database twice. The check now lives in `save_flashcard()` — the single write path shared by the review popup, *Add All*, automatic add, and Mykola's chat saves — so every route is covered at once:

- A card whose **word + part of speech** already exists anywhere in the database is skipped (word case-insensitive; the pos comparison NULL-safe, so pos-less .mht imports deduplicate too — something a database UNIQUE index could not do, since MySQL treats NULLs in a unique key as distinct).

- **The user is told:** the popup button turns into *"Already in DB"*, and the automatic-add banner counts skips — *"Added 1 card(s), skipped 1 already in the database"*.
- A **maintenance script** (`maintenance/dedup_flashcards.py`, automation #10) cleans out duplicates accumulated before the fix: dry-run by default, `--apply` deletes, always keeping the oldest card of each group. To be run once against the live database after the next deploy.

5. Quality: suite repaired, then doubled (kquantorflow_automation #6, #8, #10)

- **Suite repair** (#6): the offline app tests had been silently broken — `authlib` missing from the requirements meant 35 tests couldn't even import the app, and 4 more had gone stale against the current UI. Fixed and green again at 42 tests.
- **30 new regression tests** (#8, #10) covering the whole week's work: config-file self-creation, corrupt files surviving untouched, `POST /settings` validation and per-identity persistence, popup markup and pre-fill, auto-add behaviour, provider dispatch with every fallback path, the Bing/Oxford/M-W parsers against canned responses, duplicate-skip logic against a fake database, and the dedup script's grouping rules.
- **Result: 72 passed**, fully offline, in under 2 seconds — plus an autouse fixture that redirects the settings store into a per-test temp directory so page renders can't write into the real checkout.

6. Test ergonomics and tooling (kquantorflow_automation #5, #9)

- **`gate.py`** (#9): one shared helper to pass the site's keyword gate in any scripted session — `enter_gate()` returns an authenticated session using `ACCESS_KEYWORD` from `.env`, with a clear error when the keyword is rejected. The live tests now use it, and it doubles as a CLI smoke check. A stale live test (`favicon icon.png` → `icon.jpg`) was fixed on the way, taking the live suite from 5/6 to 6/6.
- **Presentation tooling** (#5): a reusable python-pptx deck kit (`presentation/deck_kit.py`, KuantorFlow palette, 16:9) and a stdlib-only OOXML text extractor — future decks start from a library, not from scratch.

7. Documentation and idea work (kquantorflow #95)

- **Spaced-repetition (SM-2) proposal** (refs #94): a full write-up of a review-mode design for the flashcards, rendered to `reports/` in all three formats with the report tooling (whose python-docx dependency is now pinned).
- README grew sections for the settings store, the Settings popup, the four providers, and duplicate protection as each PR landed.

Technical Highlights

- **TLS fingerprinting is the new bot wall.** The Bing investigation proved the block was neither cookies nor tokens (a full browser-session replay still 401'd) but the TLS client fingerprint itself — and found the legitimate alternative route that the Edge browser uses, which works from plain requests.
- **One write path, one check.** Because every card-saving flow funnels through `save_flashcard()`, the duplicate rule was implemented (and tested) exactly once — the popup, batch add, auto-add, and AI-chat saves all inherited it in the same commit.
- **Application-level uniqueness over a DB index — deliberately.** MySQL's unique indexes treat NULLs as distinct, so an index on (`word`, `pos`) would never catch repeated `pos-less .mht`

imports; the NULL-safe `<=>` check covers them and needed no data migration.

- **Resolve dependencies at call time for testability.** Module-level dicts capturing fetcher functions at import made monkeypatching impossible in tests (and silently hit the real network); switching the provider dispatch to call-time resolution fixed both.
- **Stacked PRs kept reviews clean:** the #21 implementation PR was based on the #105 UI branch, so each diff stayed focused while merging in order retargeted everything automatically.

Lessons Learned

1. **When replaying a request fails despite identical headers, cookies, and body — suspect the transport, not the payload.** Verifying the same request from inside the browser (200) versus Python (401) isolated TLS fingerprinting in minutes; hours could have been lost tweaking tokens.
2. **A test suite that can't import the app fails silently as "errors", not "failures"** — and can stay broken for days. The missing-authlib episode argues for running the suite (not just writing it) after every dependency change in the app repo.
3. **Fixtures that redirect filesystem side effects must arrive with the feature.** The moment the settings store learned to create files on first read, every page-rendering test began writing into the real checkout — caught only because the companion-test PR added the temp-dir fixture in the same change.
4. **HTML-escaping belongs in test expectations.** Assertions on flash messages with apostrophes must expect `'` — the rendered page never contains the raw quote.
5. **Board hygiene pays off at report time:** several Highest-Prio tickets delivered this week (#86, #13, #20, #21) are still open on the board — closing tickets when their PRs merge keeps the dashboard truthful.

Plans for Next Week

Taken from the KuantorFlow Improvements GitHub Projects board ([dashboard](https://github.com/users/Kuantor/projects/2)). The board tracks 42 items as Done. Priority order below: Highest Prio first, then Nice To Have, as requested.

Highest Prio

Repo	Issue	Title	Note
kuantorflow	#98	Add "Reset Auth" button in Settings to clear Google login and keyword cache	Next up — extends the new Settings popup
kuantorflow	#102	[Bug] Prevent unauthenticated users from modifying default settings	Next up — hardening of the new POST <code>/settings</code>
kuantorflow	#86	Add config JSON files to store application settings	Delivered this week (PRs #97, #105) — ready to close
kuantorflow	#13	Settings Menu with Auto-Add Cards Option	Delivered this week (PR #105) — ready to close
kuantorflow	#20	UI: Add Dictionary Selection Feature in Settings	Delivered this week (PR #105) — ready to close

kuantorflow	#21	Implement Dictionary Selection Feature in Settings	Delivered this week (PR #106) — ready to close
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Nice To Have

Repo	Issue	Title
kuantorflow	#54	Investigation: Android app clone of the KuantorFlow website
ai_agent	#48	Let Mykola know that he is Claude-powered
kuantorflow	#15	Parallel PRs for features and tests
kuantorflow	#87	Add the user's Google avatar to the chat widget
kuantorflow	#88	Create a docx describing how to run backup/restore
kuantorflow	#89	Invisible field for the name of the user who added a card
kuantorflow	#90	Do we tell the user that the logs are stored on the server?
kuantorflow	#92	Spaced-repetition review mode
kuantorflow	#93	Complete multi-user card ownership
kuantorflow	#99	Show message about API credits if needed
kuantorflow	#100	Quiz: treat perfective and imperfective verb answers as equal using the AI agent
ai_agent	#51	Support user-preferred nicknames in place of email login
kuantorflow	#103	Make Claude write test reports for each PR
kuantorflow	#104	Use the Allura font for the 'Welcome to KuantorFlow' message

In Progress

Repo	Issue	Title
kuantorflow	#45	UI: enhance behaviour of MHT-processing results window
kuantorflow	#55	Make config.json unique for each Gmail user
kuantorflow	#30	Create logs for various actions in the app
kuantorflow	#46	Settings: add language visibility switches
kuantorflow	#84	Darken the main page when the welcome popup is shown
kuantorflow	#85	Full-screen widget mode with visible maximize button

Todo

Repo	Issue	Title
kuantorflow	#22	Redesign website for a modern look
kuantorflow	#74	Dark theme for the website
kuantorflow	#78	Flashcards activity with Quizlet-style animation
kuantorflow	#79	Language disable checkboxes in Settings

kuantorflow	#94	Spaced-repetition review mode (SM-2)
kuantorflow	#75	About popup: agent's widget overlaps with it
kuantorflow	#76	Scheduled recaps with random log selection
kuantorflow	#25	Export/import cards (local ↔ remote database)
kuantorflow	#19	Tutorial tip above "Look Up & Save" button
kuantorflow	#56	Discussion: privacy disclosure & history bootstrap for per-user chat memory
kuantorflow	#66	Create a separate board for done items
ai_agent	#19	Teach Mykola to complement and extend the meanings of cards
ai_agent	#47	Audit issue-61 changes: remove/refactor code not needed by the embedded widget
ai_agent	#49	Runner setting for number of last logs
ai_agent	#50	Make Mykola think faster?
kuantorflow_automation	#4	Basic automated tests for the AI agent

Deployment follow-ups (carried on the tickets)

- Pull both repos on PythonAnywhere and reload (no new dependencies).
- Run `maintenance/dedup_flashcards.py` once against the live database (backup → dry run → `--apply`), per the note on #101.
- Check that Bing/Oxford/Merriam-Webster are reachable from PythonAnywhere's IPs, per the note on #21 — fallbacks keep lookups working either way.

Report generated 17 July 2026 from GitHub PR, issue, and commit data across the three repositories, and from the KuantorFlow Improvements project board. Merged-PR window: 14 July 2026 (kuantorflow#95) through 17 July 2026 (kuantorflow#107, kuantorflow_automation#10).